

Carlo Tosoni

Nationality: Italian **Date of birth:** 16/04/1999 **Place of birth:** Assisi, Italy **Gender:** Male **Phone:** (+39) 3459913191 (Mobile) **Email address:** tosonicarlo99@gmail.com **LinkedIn:** Carlo Tosoni **Github:** CarloTosoni99 **Website:** https://CarloTosoni.eu **Address:** Pisa, Italy (Home)

ABOUT ME

Master's degree in computer science and PhD candidate. Passionate software developer enthusiast in devising elegant and efficient solutions for seemingly convoluted problems.

EDUCATION & TRAINING

PhD Student in Computer Science

Ca' Foscari University of Venice [08/09/2023 - Current]

Address: Via Torino 155, 30170, Venice (Italy) | **Level in EQF:** 8

Sorting is one of the most powerful techniques for optimizing search queries. For instance, with binary search it is possible to retrieve an item from a sorted list in logarithmic time w.r.t. the list size. Sorting-based strategies can also be applied to more convoluted objects, such as finite-state automata. In fact, similarly to the Burrows-Wheeler transform, sorting the states of automata based on the strings reaching them enables the development of efficient indexes on their recognised regular language.

The PhD programme aims to significantly improve this indexing strategy by providing efficient algorithms to sort the states of automata and to run the resulting index. This research line finds applications in areas such as bioinformatics, where indexing labelled graphs is required to find patterns of interest in pangenome graphs, i.e., graphs encoding the genetic variation in a given population.

The PhD programme is funded by the following ERC grant provided by the European Union, and it is supervised by Professors Nicola Prezza and Ruben Becker at the Ca' Foscari University of Venice.

MSc in Computer Science, curriculum Big Data Technologies

University of Pisa [01/09/2021 - 01/07/2023]

Address: Largo Bruno Pontecorvo 3, 56127, Pisa (Italy) | **Final grade:** 110/110 cum laude | **Level in EQF:** 7 | **Thesis:** Compressing the Burrows-Wheeler transform of finite-state automata using run-length encoding

This master's degree, offered by the Department of Computer Science, is provided entirely in English and focuses on the following topics.

• Design, analysis and implementation of advanced efficient algorithms and data structures. Design and implementation of advanced databases. State-of-the-art techniques for the analysis of genomic datasets.
• Principles and paradigms of machine learning. Analysis and implementation of algorithms for data mining.
• Megadata analysis, search engines, and information retrieval.
• High performance computing and parallel computing.

Achieved seven times the grade 30/30 cum laude. Weighted average at the end of studies 30.47/32.

BSc in Computer and Electonic Engineering

University of Perugia [31/08/2018 - 31/08/2021]

Address: Via Goffredo Duranti 93, Perugia (Italy) | **Final grade:** 110/110 cum laude | **Level in EQF:** 6 | **Thesis:** Development of a decentralized blog using Solid technology

• Feedback control systems, Signal theory, Principles of automatic control, Internet basics.
• Algorithms and data structures, Database management systems, Programming (Java and C).

• Logic design and microcontrollers, Circuit theory, Electronic devices and technologies.

Secondary-School Degree

Liceo Scientifico Statale Annesso al Convitto Nazionale "Principe di Napoli" [09/2013 - 07/2018]

Address: Piazza Giacomo Matteotti 6706081, Assisi (Italy) | **Final grade:** 100/100

PROGRAMMING SKILLS

Programming languages

Excellent knowledge of object-oriented and imperative programming languages such as Java, C++, C#, and C. Good knowledge of Python 3 and web languages such as HTML, JavaScript and CSS. Occasional experience with Rust, MATLAB and functional programming languages such as Haskell.

Frameworks and libraries

Experience with frameworks such as React, Qwik and Node.js to develop web applications. Familiarity with the programs Unity and Blender for designing virtual realities and creating 3D models. Good knowledge of JavaFX for creating desktop applications.

Relational databases

Excellent knowledge of relational algebra and ACID database management systems. Proficient in optimizing SQL queries and working with database systems such as PostgreSQL.

Machine learning and data mining

Extensive usage of libraries such as Keras, PyTorch and scikit-learn for training and assessing neural networks and support vector machines. Knowledge of Pandas for applying data mining techniques on datasets.

SOFTWARE PROJECTS

Prefix-Free Parsing plus

(private repository) C++ implementation of an efficient and theoretical grounded version of a prefix-free parsing grammar, which computes the Burrows-Wheeler transform of repetitive genomic datasets.

Personal webpage

(public repository) Source code of my personal webpage. The project has been implemented using the framework Qwik (TypeScript), designed to generate instantly interactive web applications.

Link: https://github.com/CarloTosoni99/webpage

RESEARCH EXPERIENCES ABROAD

[01/05/2026 - 31/07/2026]

University of Florida

Visited the research group of Professor Christina Boucher at the University of Florida, in Gainesville, Florida, USA. The purpose of the mobility was to improve state-of-the-art algorithms to sequence/analyse genomic datasets by using strategies based on the Burrows-Wheeler transform.

[01/04/2025 - 30/06/2025]

University of Helsinki

Visited the Department of Computer Science of the University of Helsinki. The visit aimed to collaborate with the research group of Professor Veli Mäkinen on topics concerning compressibility measures for tries.

PUBLICATIONS

[2026]

Faster Cache-Efficient Pattern Matching for Deterministic Wheeler Pangenome Graph

Authors: Riccardo Maso, Nicola Prezza, and Carlo Tosoni

Italian Conference on Theoretical Computer Science ICTCS

[2025]

New Entropy Measures for Tries with Applications to the XBWT

Authors: Lorenzo Carfagna and Carlo Tosoni

arXiv, submitted to the Journal of Computer and System Sciences JCSS

[2025]

Analysing New Entropy Measures for Tries

Authors: Lorenzo Carfagna and Carlo Tosoni

International Symposium on String Processing and Information Retrieval SPIRE

[2025]

Encoding Co-Lex Orders of Finite-State Automata in Linear Space

Authors: Ruben Becker, Nicola Cotumaccio, Sung-Hwan Kim, Nicola Prezza, and Carlo Tosoni

Annual Symposium on Combinatorial Pattern Matching CPM

[2024]

Indexing Finite-State Automata Using Forward-Stable Partitions

Authors: Ruben Becker, Sung-Hwan Kim, Nicola Prezza, and Carlo Tosoni

International Symposium on String Processing and Information Retrieval SPIRE

CERTIFICATIONS

[British Council, 19/03/2022]

IELTS Academic 7.0 (CEFR C1), Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0. The International English Language Testing System (IELTS) is an international standardised test of English language proficiency for non-native English language speakers provided by the British Council.

[Huawei Technologies Co. Ltd, 24/01/2021]

Huawei HCIA Routing and Switching Certification

TEACHING ACTIVITIES

[01/03/2026 - 31/05/2026]

Tutor at the Ca' Foscari University of Venice

Appointed tutor for the master's course "Algorithms and Learning over Massive Data" (a.y. 2025/26) and the summer school SEAA "Scuola Estiva di Algoritmi Avanzati" (a.y. 2023/24).



Pisa, 16/10/2026

Carlo Tosoni